THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 9: **Some Applications of Trigonometry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The angle of elevation of the top of a vertical tower from a point on level ground is 30°. After walking 4 m towards the foot of the tower along a straight line, the angle of elevation becomes 45°. The height of the tower is: 1

(a) $\frac{4\sqrt{3}}{3}$ m **(b)** 4 m **(c)** $2+2\sqrt{3}$ m **(d)** $6+2\sqrt{3}$ m

2. A flagstaff stands on the top of a building 10 m high. From a point on level ground the angles of elevation of the bottom and the top of the flagstaff are 30° and 60° respectively. The length of the flagstaff is: 1

(a) 30 m **(b)** 40 m **(c)** $20\sqrt{3}$ m **(d)** 20 m

3. A car moves at a uniform speed towards the foot of a vertical tower. The angle of elevation of the top of the tower changes from 30° to 60° in 4 minutes. The further time the car takes to reach the foot of the tower is: 1

(a) 2 minutes **(b)** 8 minutes **(c)** $\frac{4}{3}$ minutes **(d)** 4 minutes

4. From the top of a tower $6\sqrt{3}$ m high, the angles of depression of two objects on level ground, on the same side of the tower and in line with its foot, are 30° and 60°. The distance between the two objects is: 1

(a) 12 m **(b)** 6 m **(c)** 24 m **(d)** $12\sqrt{3}$ m

5. From the top of a building 3 m high, the angle of elevation of the top of a tower is 30° and the angle of depression of its foot is 60°. The height of the tower is: 1

(a) $4\sqrt{3}$ m **(b)** 4 m **(c)** 1 m **(d)** $\sqrt{3}$ m

6. Two points on level ground lie on opposite sides of a tower of height 9 m, in line with its foot. The angles of elevation of the top of the tower from the two points are 45° and 60°. The distance between the two points is: 1

(a) $9+3\sqrt{3}$ m **(b)** 12 m **(c)** $9-3\sqrt{3}$ m **(d)** $9+9\sqrt{3}$ m

7. A person looks at the top of a vertical pole from a point on the ground. The top of the pole is $4\sqrt{3}$ m above the level of the person's eyes and the eyes are 12 m horizontally from the pole. The angle of elevation of the top of the pole is: 1

(a) $30\sqrt{3}^\circ$ **(b)** 45° **(c)** 30° **(d)** 60°

8. The angle of elevation of the top of a vertical tower from a point on level ground is 30°. After walking 6 m towards the foot of the tower along a straight line, the angle of elevation becomes 45°. The height of the tower is: 1

(a) $2\sqrt{3}$ m **(b)** 6 m **(c)** $3+3\sqrt{3}$ m **(d)** $9+3\sqrt{3}$ m

9. **Assertion (A):** When an observer walks towards the foot of a tower, the angle of elevation of its top increases. 1

Reason (R): For a tower of fixed height h at horizontal distance d, $\tan\theta = \frac{h}{d}$, which increases as d decreases.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** A tower $10\sqrt{3}$ m high casts a shadow 10 m long when the angle of elevation of the Sun is 60°. 1

Reason (R): $\tan 60^\circ = \frac{1}{\sqrt{3}}$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. The angle of elevation of the top of a vertical tower from a point on level ground is 45°. After walking 6 m towards the foot of the tower along a straight line, the angle of elevation becomes 60°. Find the height of the tower. **2**

12. A flagstaff stands on the top of a building 9 m high. From a point on level ground the angles of elevation of the bottom and the top of the flagstaff are 45° and 30° respectively. Find the length of the flagstaff. **2**

13. A car moves at a uniform speed towards the foot of a vertical tower. The angle of elevation of the top of the tower changes from 30° to 60° in 6 minutes. Find the further time the car takes to reach the foot of the tower. **2**

14. The angle of elevation of the top of a vertical tower from a point on level ground is 45°. After walking 12 m towards the foot of the tower along a straight line, the angle of elevation becomes 60°. Find the height of the tower. **2**

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A building is $18\sqrt{3}$ m high. From the top of the building the angle of elevation of the top of a tower is 30° and from the foot of the building it is 60°. Find the height of the tower and its distance from the building (in this order). **3**

16. A flagstaff stands on the top of a building 14 m high. From a point on level ground the angles of elevation of the bottom and the top of the flagstaff are 45° and 60° respectively. Find the length of the flagstaff. **3**

17. Two points on level ground lie on opposite sides of a tower of height 6 m, in line with its foot. The angles of elevation of the top of the tower from the two points are 45° and 60°. Find the distance between the two points. **3**

OR

A building is $20\sqrt{3}$ m high. From the top of the building the angle of elevation of the top of a tower is 30° and from the foot of the building it is 60°. Find the height of the tower and its distance from the building (in this order). **3**

18. A flagstaff stands on the top of a building 16 m high. From a point on level ground the angles of elevation of the bottom and the top of the flagstaff are 45° and 60° respectively. Find the length of the flagstaff. **3**

OR

Two points on level ground lie on opposite sides of a tower of height 12 m, in line with its foot. The angles of elevation of the top of the tower from the two points are 45° and 60°. Find the distance between the two points. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A vertical tower stands on level ground. From a point P on the ground, 20 m from the foot of the tower, the angle of elevation of its top is 60°. (Use $\sqrt{3} = 1.73$.) **5**

(i) Name the trigonometric ratio that relates the height of the tower to the distance of P from the foot of the tower.
(ii) Write the value of $\tan 60^\circ$.
(iii) Find the height of the tower.
(iv) The observer moves back until the angle of elevation becomes 30°. Find the new distance from the foot of the tower and how far the observer moved.

20. A building is 6 m high. From the top of the building the angle of elevation of the top of a tower is 45° and from the foot of the building it is 60°. Find the height of the tower and its distance from the building (in this order). **5**

OR

Two vertical poles of equal height stand on opposite sides of a road 72 m wide. From a point on the road between them, the angles of elevation of the tops of the poles are 60° and 30°. Find the common height of the poles and the distances of the point from the poles (nearer pole first). **5**

END OF QUESTION PAPER — Total: 40 Marks